

Rex Gonzalez

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EDUCATION

University of South Florida

B.S in Computer Science, Minor in Mathematics. Expected Graduation: December 2026

Tampa, FL

Aug 2023 – Present

TECHNICAL SKILLS

Programming Languages: Python, C/C++, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks & Libraries: TensorFlow, PyTorch, Nest, React, Node, Flask, Numpy, OpenCV, Streamlit

Databases: PostgreSQL, MongoDB, MySQL

Developer Tools: Docker, Git, GitHub, Google Cloud Platform, Visual Studio Code, AWS, Conda

EXPERIENCE

Researcher at Reality, Autonomy, and Robot Experience (RARE) Lab

April 2025 – Present

University of South Florida under Professor Zhao Han

Tampa, FL

- Conducting research in Human-Robot Interaction (HRI) and AI/ML for socially assistive robots to improve and enable new autonomous interaction systems.
- Integrated hardware and software to address real-world challenges in HRI, robotics, and software engineering.
- Collaborated with students and faculty to design and prototype **Bloom**, a socially assistive robot for elderly care.

Community Lead at Google Developer Group (GDG)

June 2025 – Present

University of South Florida

Tampa, FL

- Lead a tech organization with 800+ members, offering workshops, technical talks, mentorship, and HackUSF.
- Coordinate outreach to industry professionals for sponsorships and speaking at HackUSF and GDG initiatives.
- Collaborate with student organizations across campus to co-host engaging events and skill-building workshops.

PROJECTS

Bloom: A Low Cost Social Robot for the Elderly | *Python, C++, AI, STT, TTS, RAG* April 2025 - Present

- Designed and prototyped **Bloom**, a \$300 social robot for the elderly, an improvement of Cornell's Blossom robot.
- Integrated speech-to-text, text-to-speech, touch sensing, face recognition, and RAG to enable natural interaction.
- Designed an exploratory non-pharmacological treatment for loneliness, Alzheimer's, and related dementias in collaboration with the USF College of Public Health and Aging Studies.

USF AI Chatbot for Academic Planning | *Python, Javascript, RAG, Webscraping*

April 2025

- Developed a RAG AI advisor, improving the efficiency of class searches by 40%, answering students' questions and providing suggestions regarding their situation.
- Implemented an interactive flowchart that is connected to the registration database and *Rate My Professors* that allows students to see key information about classes and professors.
- Developed personalised flowcharts by gathering student information, wants, and needs. Students could create an account and save their information, flowcharts, and past chats.

Cognia: EEG Assistive Systems and Neurotech | *Python, C++, Arduino, BCI, WebSocket, RAG* October 2025

- Utilized the Emotiv Insight EEG headset to get mental commands and EEG metrics to control movement and collect mental health insights, which **won the Assistive Technology Track** at HealthHacks 2025.
- Implemented real-time Python-Arduino communication via WebSockets and serial protocol to power mechanisms for users with motor impairments.
- Built an AI dashboard analyzing EEG metrics and the users' qualitative data, giving mental/physical health insights and allowing the user to have a conversation about their results.

ScheduleSync | *Python, HTML/CSS, JavaScript, Webscraping*

October 2025 - Present

- Built a web-based scheduling tool that identifies availability across users' calendar files to schedule meetings.
- Developed backend logic in Python to parse, merge, and normalize **.ics** files. Implemented robust timezone handling, input validation, and algorithms to compute mutual availability slots of 30+ minutes.
- Integrated **BeautifulSoup** to scrape HTML-based university schedules and convert them into calendar events and ensured accurate data flow and compatibility with Google Calendar and Outlook.